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RESEARCH-ARTICLE

Psychometric Properties of the User Experience Questionnaire (UEQ)

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ABSTRACT

User experience (UX) summarizes user perceptions and responses resulting from the interaction with a product, system, or service. The User Experience Questionnaire (UEQ) is one standardized instrument for measuring UX. With six scales, it identifies areas in which product improvements will have the highest impact. In this paper, we evaluate the reliability and validity of this questionnaire. The data of $N = 1,121$ participants who interacted with one of 23 products indicated an acceptable to good reliability of all scales. The results show, however, that the scales were not independent of each other. Combining perspicuity, efficiency, and dependability to pragmatic aspects as well as novelty and stimulation to hedonic aspects of UX improved the model fit significantly. The systematic variations of product properties and correlations with the System Usability Scale (SUS) in a second experiment with $N=499$ participants supported the validity of these two factors. Practical implications of the results are discussed.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *HCI theory, concepts and models*.

KEYWORDS

user experience, psychometric properties, UEQ

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1 INTRODUCTION

Research on user experience (UX) began more than 20 years ago, around the turn of the millennium, with the argument that usability research was too focused on task efficiency and work. The goal was to design *pleasurable products* [22], to increase *hedonic quality* [20], or to *engineer joy* [18]. UX has been supposed to complement the task-oriented view of the usability of interactive products by hedonic qualities, such as aesthetics [33], self-actualization [16], and positive emotions [21] that people experience when interacting with a product. It goes beyond goal achievements. Interacting with products also fulfills the human needs for stimulation and personal growth [17], evokes memories [26], and communicates messages about self-identity in social settings [2]. After a period of defining and conceptualization (see [21] for a review), UX is now defined as "*a person's perceptions and responses that result from the use or anticipated use of a product, system or service.*" (ISO 9241-210 [13]).

UX describes the subjective (cognitive and affective), evaluative judgment of a product by the user, i.e., it results from the interaction between characteristics and needs of the user with the properties of the product. Accordingly, even when using and interacting with the same product, users' experience might strongly vary, depending on the heterogeneity of their current needs and previous experiences.

There are two main approaches to measure UX. Either subjective statements of the users are gathered, e.g., by (standardized) questionnaires or interviews, or UX is inferred from objective measures, e.g., gaze direction, response times or physiological parameters [1]. Both approaches allow a complementary view on different aspects of UX. In the current paper, we focus on standardized questionnaires. They can be applied, analyzed, and interpreted fast and easily. Thus, it is very attractive to use them in user studies. Several standardized questionnaires aim at measuring UX, e.g., the Questionnaire for User Interaction Satisfaction (QUIS) [14], the Standardized User Experience Percentile Rank Questionnaire (SUPR-Q) [28], the AttrakDiff2 [19], and the User Experience Questionnaire (UEQ) [23].

When standardized questionnaires are considered to be applied in a user study, they need to fulfill at least three main quality criteria (i.e., psychometric properties of the test) – objectivity, reliability, and validity.

- *Objectivity* is given when measurements are independent of effects of the experimenter or situational conditions during application, analysis, and interpretation.

- *Reliability* is an indicator for the formal (or statistical) accuracy of a scientific measure. High reliability means that the same results can be reproduced each time a questionnaire is applied within the same study setup.
- *Validity* means that a concept, conclusion, or measurement correspond to the real world, i.e., whether a questionnaire measures that construct which it intends to measure.

The User Experience questionnaire (UEQ) has been widely used for more than 10 years now. It has been translated into different languages and applied in different scenarios (see e.g. [29]). However, only a few studies reported about its reliability and validity so far [23, 27] (see also below).

The contribution of this paper is a psychometric assessment of the UEQ. In particular, we provide the first assessment of the underlying theoretical model with a factor analysis on the basis of a large sample ($N = 1,121$) with 23 different products (*construct validity*). Second, we correlated the scales of the UEQ with a measurement of usability (the System Usability Scale, SUS) to provide evidence for *divergent and convergent validity*. Third, we report *reliability* measures for each scale for different products to estimate the reliability of the test. Finally, we run a validation study with $N = 499$ participants. We manipulated product features experimentally to measure the effect on particular scales of the UEQ in order to provide further evidence for the validity of the questionnaire (*validity based on external criteria*). This research has implications for (a) understanding the advantages and limitations of applying the UEQ in user studies and, thus, (b) give interested researchers and potential clients the opportunity to decide whether this questionnaire suits their needs in their evaluation project. This paper does not intend to revise the UEQ or its underlying theoretical model.

1.1 User Experience Questionnaire (UEQ)

The UEQ has been constructed by Laugwitz and colleagues [23]. It is based on the theoretical model proposed by Hassenzahl [15]. This model distinguishes between perceived ergonomic quality, perceived hedonic quality, and perceived attractiveness of a product. Perceived ergonomic quality and perceived hedonic quality describe independent dimensions of UX. Ergonomic quality is goal- or task-oriented, i.e., high ergonomic quality enables the user to reach his or her goals efficiently and effectively. In this sense, ergonomic quality corresponds to usability. In contrast, hedonic quality is non-task oriented and describes the originality of the design or the beauty of the user interface [23]. Perceived attractiveness represents an overall measurement of the product in terms of like / dislike.

In its final version, the UEQ consists of 26 bipolar items in form of a semantic differential, e.g., predictable / unpredictable, fast / slow, or boring / exciting, with seven options to choose between these endpoints. The items are assigned to six scales:

- *Attractiveness* measures the overall impression of the product in terms of like / dislike.
- *Perspicuity* measures how easy it is to get familiar with the product.
- *Efficiency* measures whether users can solve their tasks without unnecessary effort.
- *Dependability* measures whether users feel in control of the interaction.

- *Stimulation* measures how exciting and motivating it is to use the product.
- *Novelty* measures whether the product is innovative, creative, and interesting for the user.

Attractiveness is supposed to be a pure valence dimension, giving an overall evaluation of the product in terms of like/dislike. *Perspicuity*, *Efficiency*, and *Dependability* are pragmatic quality aspects (goal-oriented), whereas *Stimulation* and *Novelty* measure hedonic quality aspects (not goal-oriented). With only 26 items, the UEQ is a very efficient measurement instrument. It can be used to compare two products, but also to benchmark the current status of the product [23] and it allows “at least educated guesses about the areas where improvements will have the highest impact” (p.6).

A manual describes in detail how to apply the questionnaire and how to interpret the results. An analysis tool is presented that makes data analysis easy and minimizes errors. In this sense, an *objective* measurement of UX is possible. However, not much is known about the *reliability* and *validity* of the questionnaire.

1.2 Related Work

Reliability and validity of the UEQ were first evaluated during test construction. Laugwitz et al. [23] reported reliability estimates based on internal consistency (Cronbach’s α) between $\alpha = .65$ (scale *dependability*) and $\alpha = .89$ (scale *attractiveness*) for the German version of the UEQ. For the English version, the reliability ranged between $\alpha = .69$ (scale *dependability*) and $\alpha = .88$ (scale *stimulation*) [23]. However, their sample was very small with $N = 16$ or $N = 21$, respectively. Similar reliability scores were reported for the Spanish UEQ [27]. Cronbach’s alpha ranged from $\alpha = .48$ (scale *dependability*) to $\alpha = .85$ (scale *attractiveness*) in a study with $N = 94$ students and from $\alpha = .55$ (scale *dependability*) to $\alpha = .83$ (scale *attractiveness*) in a study with $N = 95$ students.

Laugwitz et al. [23] conducted two studies to evaluate the validity of the questionnaire. In a first study with $N = 13$ participants, the three scales of pragmatic quality correlated negatively with task-completion time, i.e., participants were faster with more usable products. As expected, the two scales of hedonic quality did not correlate with task-completion time. Discriminant and convergent validity were assessed in a second study with $N = 16$ participants. The authors correlated the scales of the UEQ with those of the AttrakDiff. Both questionnaires are supposed to measure similar aspects of UX. However, both questionnaires also overlap in about 40% of the items. Thus, the observed correlations were not surprising. Also, the sample size was very small in both studies.

Finally, Rauschenberger et al. [27] have reported that “the reliability (i.e. the scales are consistent) and validity (i.e. the scales do really measure what they intend to measure) of the UEQ scales was investigated in several studies (in 11 usability tests with a total number of 144 participants and an online survey with 722 participants). A review of all available studies showed that reliability (Cronbach’s Alpha was used for an estimation of internal consistency) of the scales was sufficiently high.” However, to our knowledge, the results of these studies were not published in detail.

2 STUDY 1: EVALUATION

User experience is a result of the interaction between a person (with all its characteristics) and the properties of the product. Many user studies have a lack of ecological validity. They evaluate a single product or prototype in a laboratory environment after a short interaction time. However, observed effects (e.g., correlations between sub-scales) heavily depend on the stimulus material provided. In order to evaluate the UEQ with regard to its criteria of good quality, it needs to be filled in not only by a variety of different people but it also needs to be applied to a variety of different products. Thus, we combined the results of smaller user studies that were originally run to evaluate a particular product after the design process or for research purposes in order to re-analyzed the data. Products were selected that maximize the variance within in data. In summary, we present the results of 1,121 participants who evaluated one of 23 different products (see Table 1 and below) in laboratory, in field, or online.

2.1 Methods

Participants. A total of 1,184 participants volunteered in the studies, after they had given their written informed consent. Participants were recruited by convenience sampling. No reward was given. Participants with missing data (i.e., not all questions of the UEQ have been answered) were excluded. Thus, 1,121 participants remained in the sample (896 male, 387 female (38 missing values) with an age range of 18-78 years with two peaks (20-27 years and 52-57 years)). About one third hold a university degree. About 39% had a technical or natural-scientific background, the others worked in commercial (11%), medical (3%), juristic (1%), pedagogic (4%), administrative (1%), manual (2%) or artistic (3%) occupational areas (about 34% did not indicate occupation). That is, the sample was heterogeneous with regard to age and education. Importantly, participants were no experts in UX. All studies were carried out in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki.

Product Selection Process. User studies were included that fulfilled the following inclusion criteria. A user study should include at least 20 participants. Participants interacted with the product for at least 5 minutes. In all studies, participants were introduced to the particular scenario and solved particular tasks. They had filled in the UEQ and maybe further usability questionnaires (e.g., the SUS). There should be tasks included that participants were familiar with (e.g., using a search engine) and those that were new and innovative (e.g., selecting an object in a room with augmented reality). There should be studies included with different devices (e.g., computer, smartphone, wearable), interfaces (e.g., website, browser, IDE), and locations (in laboratory, in field, online).

Evaluated Products. The selected products can be divided into six classes. The IDs refer to Table 1.

Authentication process (ID 1-6). Publicly accessible Wi-Fi hotspots are used, for example, for corporate guest networks. Six different processes for token- and context-based authentication in public Wi-Fi infrastructures were evaluated with regard to their usability and perceived security (i.e., Two-Dimensional Signal Transmission Sheet (2DST), audio fingerprint (AudioContext), Microsoft Kinect to identify a login attempt, Near Field Communication (NFC), password, and QR code; for details, see [5, 6]).

Automatic and manual adaption of font size (ID 7-11). Different prototypes of a context-sensitive system were evaluated with regard to their usability and user experience. The font size of the browser adapted to the activity of the user (e.g., walking, standing, sitting), either automatically or manually.

Search engines (ID 12-14). Participants assessed one of three different search engines (Google, Ecosia, and Startpage¹). These search engines were chosen because they were similar in their user interface but different in data security, the quality and presentation of search results, and sustainability. Participants were asked to find three different types of information each (*facts*, e.g., "how many people did live in Munich in 1980", *translations*, e.g., "what is the Finnish word for "mountain top", or a *picture*, e.g. an analog clock showing half past five). The tasks were the same for all search engines. This study was run online with a survey tool.

Object selection in a smart home environment (ID 15-18). In this scenario, participants were asked to find and select a particular object (e.g., a lamp) among few or many other objects (e.g., TV, stereo system, fan). For this purpose, participants used a smartphone, either with an augmented reality application or with a simple list of objects.

Finding a location with an interactive map (ID 19-22). Participants were asked to find a particular location (i.e., a lecture hall on the campus of the university), either with the website of the university (which includes an interactive campus map²) or with Google Maps³. They either used a laptop or a mobile device and rated the maps with regard to their usability and user experience afterward.

Programming environment (ID 23). Students without prior experience of MATLAB⁴ (a commercial software for solving mathematical problems and for graphical presentation of the results) evaluated the software as an IDE after solving some predefined tasks (e.g., define and print the function $f(x) = x^3 - 3x^2 - x + 3$ in the interval $[-4.5, 4.5]$).

Further Material. In addition to the German version of the UEQ, the German translation of the System usability scale (SUS) [4] was applied in some user studies. The SUS is a standardized questionnaire with 10 short questions that primarily measure the usability aspect of a product. The SUS was used to assess the convergent and divergent validity of the UEQ scales.

2.2 Factor Structure (Construct Validity)

The factor structure is related to the construct validity of the questionnaire. It is defined as the degree to which a test measures what it claims or purports to be measuring [12]. According to Laugwitz et al. [23], the UEQ measures five independent aspects of UX (i.e., stimulation, novelty, efficiency, dependability, and perspicuity) and an overall assessment (i.e., attractiveness).

We assessed the factor structure in the following three steps:

- (1) Assessing the inter-correlations of the scales
- (2) Replicating the factor structure as described in [23]
- (3) Exploring an alternative factor structure

¹<https://www.google.com>, <https://www.ecosia.org/>, <https://www.startpage.com>

²<https://www.kit.edu/campusplan/>

³<https://www.google.com/maps>

⁴<https://www.mathworks.com/products/matlab.html>

Table 1: Overview of the rated products, data acquisition method, number of participants, and results of the UEQ and SUS.

ID	Product Name	Product Type	Product Area	Data Acq.	N	SUS	Att	UEQ Score			UEQ Reliability			Overall	
								Pragmatic			Hedonic			Pragmatic	
								Eff	Per	Dep	Sit	Nov	Att	Prag.	Hed.
1	2DST	Mobile App	Authentication	lab	21	82	1.5	1.6	2.3	1.2	1.2	1.7	.65	.88	.66
2	AudioContext	Mobile App	Authentication	lab	19	59	-0.2	-0.4	1.4	-0.2	0.3	2.2	.47	.56	.01
3	Kinect	Mobile App	Authentication	lab	20	69	0.9	1.1	1.8	0.3	1.6	2.2	.58	.63	.59
4	NFC	Mobile App	Authentication	lab	21	91	2.1	2.3	2.7	1.9	1.7	1.9	.83	.66	.65
5	Password	Mobile App	Authentication	lab	22	66	-0.9	0.0	1.6	1.2	-1.3	-2.5	.66	.76	.35
6	QR Code	Mobile App	Authentication	lab	21	78	0.9	1.0	1.8	1.2	0.6	0.8	.65	.65	.73
7	Font size auto 1	Browser Tool	Context Rec.	lab	20		1.1	0.9	1.6	0.8	1.0	1.0	.35	.83	.74
8	Font size auto 2	Browser Tool	Context Rec.	lab	24		-0.3	0.1	0.8	-0.5	-0.0	0.3	.15	.65	.60
9	Font size man 1	Browser Tool	Context Rec.	lab	20		0.8	0.9	2.0	1.3	0.4	-0.2	.71	.86	.42
10	Font size man 2	Browser Tool	Context Rec.	lab	24		0.8	0.9	1.7	1.3	0.4	-0.1	.49	.84	.81
11	Font size both	Browser Tool	Context Rec.	lab	24		0.0	0.3	1.1	0.0	0.4	0.7	.08	.81	.72
12	Google	Website	Search Eng.	online	50		1.2	1.8	2.1	1.7	0.6	0.9	.76	.81	.81
13	Ecosia	Website	Search Eng.	online	49		1.1	1.3	1.6	1.1	0.6	0.9	.81	.78	.72
14	Startuppage	Website	Search Eng.	online	49		0.2	0.5	1.2	0.7	-0.2	-0.1	.76	.79	.76
15	AR many obj.	Mobile App	Smart Home	lab	22		1.1	0.6	1.7	0.3	1.4	1.6	.63	.69	.83
16	AR few obj.	Mobile App	Smart Home	lab	21		0.9	0.6	1.8	0.2	1.5	1.6	.79	.87	.78
17	List many obj.	Mobile App	Smart Home	lab	22		0.8	1.4	1.8	1.3	-0.1	-0.1	.89	.86	.83
18	List few obj.	Mobile App	Smart Home	lab	22		0.8	1.8	2.2	1.6	-0.4	-0.4	.78	.63	.79
19	KIT	Website	Routing	field	151	63	0.3	0.4	0.7	0.7	-0.1	-0.6	.80	.88	.80
20	KIT	Website	Routing	field	148	54	-0.2	-0.0	0.1	0.3	-0.4	-0.6	.84	.88	.80
21	Google Maps	Website	Routing	field	154	85	1.6	1.9	2.2	1.7	0.8	0.1	.67	.80	.58
22	Google Maps	Website	Routing	field	154	82	1.2	1.4	1.9	1.4	0.6	0.1	.83	.89	.71
23	Matlab	IDE	Programming	lab	43		0.2	1.0	-0.2	0.6	0.5	-0.4	.61	.84	.66
Overall					1121		0.7	1.0	1.4	1.0	0.4	0.1	.80	.89	.77
Mean					49		0.7	0.9	1.6	0.9	0.5	0.5	.64	.78	.67
													.83	.85	.78
													.85	.92	.88
													.89	.92	.85
													.89	.92	.88
													.81	.87	.83

Table 2: Inter-correlations of the UEQ scales (Study 1).

	Att.	Per.	Eff.	Dep.	Sti.	Nov.
Attractiveness (Att.)	–	.72	.79	.71	.73	.47
Perspicuity (Per.)	.72	–	.74	.72	.45	.30
Efficiency (Eff.)	.87	.66	–	.73	.53	.30
Dependability (Dep.)	.61	.63	.81	–	.38	.10
Stimulation (Sti.)	.75	.40	.45	.02	–	.63
Novelty (Nov.)	.53	.36	.21	-.24	.84	–

Note: The upper-right half shows the correlation based on raw data ($N = 1,121$), the lower-left half those of the means across the 23 studies ($N = 23$). As expected, attractiveness was correlated with all other scales. But there were also great correlations between the pragmatic (green) and hedonic (orange) scales. For overall correlation, coefficients greater than $r = .10$ are statistically significant, for correlations across means, coefficients greater than $r = .30$ are statistically significant.

Inter-correlations. Laugwitz et al. [23] proposed that the scales of the UEQ were statistically independent of each other (except for attractiveness). This assumption allows conclusions about the areas where improvements will have the highest impact. It should be noted that the authors did not claim independence of the scales explicitly in their paper. However, they applied a varimax rotation after factor extraction during construction of the questionnaire – a method to reach maximum independence of factors to improve interpretability of the resulting scales. We computed correlations between the six scales of the UEQ (Pearson correlation), once across the whole sample ($N = 1,121$) and once based on the mean scores of the studies ($N = 23$). The later analysis considers unequal sample sizes across studies. For example, the overall analysis overestimates the actual correlation between two scales if the correlation was observed for a product with a large sample size. The results are shown in Table 2.

For the overall analysis, correlations of $r > .10$ reached statistical significance due to the large sample size. However, such small correlations do not have theoretical impact (in this case, the two scales have only 1% of variance in common). Thus, following the convention by Cohen [11], who proposed correlations of $r = .10$ as small, $r = .30$ as medium, and $r = .50$ as large, we only interpreted correlations of $r > .30$ as being practically significant. For correlations across mean scores, correlations of $r > .30$ were statistically significant.

As proposed by Laugwitz et al. [23], the scale *attractiveness* correlated with all further scales (all $r > .46$). Although the scales are supposed to be independent of each other [23], we observed significant correlations between some of them (see Table 2), in particular between perspicuity, efficiency, and dependability (about 40–66% common variance), as well as between novelty and stimulation (about 40–71% common variance). That is, scales measuring pragmatic aspects of a product were highly correlated as well as those scales measuring hedonic aspects.

Replication of the factor structure. In order to replicate the proposed factor structure, we run a principal component analysis (PCA)[23]. The basic assumption of factor analysis is that for a

collection of observed variables (i.e., items of a questionnaire) there are a set of underlying or latent variables called factors (smaller than the number of observed variables), that can explain the interrelationships among those items. As [23], we included all items except for those of the scale *attractiveness*, which is supposed to be a second order factor. Data quality was sufficient to perform a PCA (Kaiser-Meyer-Olkin measure of sample adequacy $KMO=.936$; Bartlett test: $\chi^2 = 12790, df = 190, p < .001$). Because the questionnaire consists of five scales, we restricted the solution to an extraction of five components. By applying a varimax rotation, a maximum independence of the factors was obtained. We expected that items of a scale load on the same component [23]. The extracted factors explained about 69% of the total variance, ranging from 8% to 16% per factor (after varimax rotation). Most of the items loaded on the expected factors (see Table 3), confirming the initial model. However, two items (Item 23 and Item 11) loaded on different factors than expected, and 4 items (Items 8, 13, 19, and 3) showed minor loadings on a second factor. It remains speculative at this point why this was the case. We hypothesize that the meaning of items were semantically ambiguous. For example, Item 23 ("organized") was supposed to reflect *efficiency* of use, i.e. whether users can solve their tasks without unnecessary effort. However, an "organized" user interface may also affect how easy it is to get familiar with a product (i.e., *perspicuity*). Similarly, Item 11 ("supportive") may not (only) describe whether users feel in control of the interaction (i.e., *dependability*) but also the effort needed to solve a task (i.e., *efficiency*). Finally, we evaluated the eigenvalues of the factor solution. The eigenvalues represent the total amount of variance that can be explained by a given principal component. Before rotation, it can be used to decide about the number of components to be extracted. The eigenvalues of two factors were below 1, suggesting that a three factor solution would have been more appropriate (Kaiser-Guttman criterion). The scree test [10] (Figure 1) even suggested a two factor solution.

Exploration of an alternative factor structure. One goal of the factor analysis is reduction of variables. From a theoretical perspective, the obtained factors need to be interpretable, e.g., as sub-scales of a questionnaire. If items load on different factors, the interpretation is not unequivocally. Because of this limitations, a

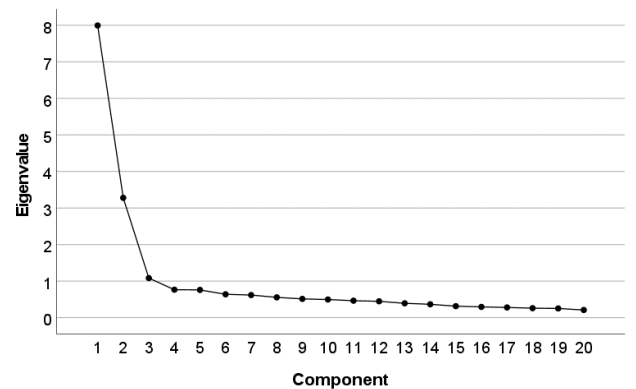
**Figure 1: Scree plot of the PCA (Study 1).**

Table 3: Factor loadings of the rotated component matrix (Study 1).

Item No.	Description	Orig. Scale	5-Factor with Varimax					2-Factor with Varimax	
			Pers.	Eff.	Dep.	Stim.	Nov.	Pragmatic	Hedonic
Item 2	understandable	Pers.	.715					.779	
Item 4	easy to learn	Pers.	.703					.717	
Item 13	easy	Pers.	.701	.441				.769	
Item 21	clear	Pers.	.724					.764	
Item 9	fast	Eff.		.743				.704	
Item 20	efficient	Eff.		.748				.786	
Item 22	practical	Eff.		.683				.664	
Item 23	organized	Eff.	.735					.566	
Item 8	predictable	Dep.		.420	.576			.708	
Item 17	secure	Dep.			.812			.612	
Item 19	meet expectations	Dep.	.530		.418			.738	
Item 11	supportive	Dep.		.727				.760	
Item 5	valuable	Stim.				.618		.517	.454
Item 6	exciting	Stim.				.764			.773
Item 7	interesting	Stim.				.766			.737
Item 18	motivating	Stim.				.664			.643
Item 3	creative	Nov.				.419	.660		.775
Item 10	inventive	Nov.					.810		.800
Item 15	leading edge	Nov.					.826		.823
Item 26	innovative	Nov.					.804		.712
Eigenvalue			3.276	3.399	1.659	2.613	2.937	6.647	4.629
% Variance explained			16.378	16.997	8.297	13.065	14.683	33.233	23.143

Note: Items are sorted by the original scales, item numbers represent the order in the questionnaire. Unexpected item loadings are highlighted; factor loadings $< .400$ are omitted; Pers.: Perspicuity, Dep: Dependability, Stim.: Stimulation, Nov.: Novelty

second factor analysis was run (PCA, Varimax rotation), which was restricted to the two factors with the largest eigenvalues. This procedure is also in line with theoretical models, which distinguish between ergonomic (or pragmatic) and hedonic aspects of product quality.⁵ The results are summarized in Table 3. The two-factor solution still explained about 56% of the total variance. Items assigned to these factors indicated that they represent hedonic or pragmatic quality aspects of UX, respectively. Except for Item 5 ("valuable") all items loaded on one of the factors. As expected by the theoretical model, attractiveness correlated with hedonic, $r = .65$, $p < .001$, and pragmatic, $r = .82$, $p < .001$, aspects of UX as well. The correlation between hedonic and pragmatic aspects was rather small but significant, $r = .42$, $p < .001$ ⁶

2.3 Reliability

The reliability of a scale describes how precise the intended construct is measured. It can be estimated, for example, by computing the internal consistency of a scale (Cronbach's $\alpha = n * r / (1 + (n - 1) * r)$, where r is the mean correlation of the items in a scale and

n is the number of items in a scale), i.e., it indicates that all items in a scale measure a similar construct. We computed Cronbach's α for each scale, separately for each product and overall (i.e., with $N = 1,121$ data sets), both reported in Table 1. The reliability of the scales varied highly between scales and across studies, which was partially due to varying sample sizes. Therefore, the best estimate of the reliability is the overall value (i.e., when all data sets were included). The reliability ranged between $\alpha = .77$ (scale *dependability*) and $\alpha = .92$ (scale *attractiveness*) (Table 1). These results replicate previous studies. For example, Laugwitz et al. [23] reported Cronbach's α between $\alpha = .65$ (scale *dependability*) and $\alpha = .89$ (scale *attractiveness*). The reliability of the scales pragmatic and hedonic quality is good with Cronbach's $\alpha = .92$ or $\alpha = .88$, respectively. Overall, the reliability is acceptable ($\alpha > .70$) to good ($\alpha > .80$).

2.4 Convergent and Discriminant Validity

A user perceives pragmatic quality when a product supports him in solving a task effectively and efficiently. In this sense, pragmatic quality is related to usability. If this were the case, aspects of pragmatic quality (i.e., the UEQ scales *efficiency*, *dependability*, and *perspicuity*) should correlate with the SUS (convergent validity). In contrast, hedonic quality reflects the user's joy and fun when using the product. Thus, aspects of hedonic quality (i.e., the UEQ scales *stimulation* and *novelty*) should not correlate with the SUS (discriminant validity). Because attractiveness is a common quality that

⁵We also performed a factor analysis with a three-factor solution. Here, the items of the pragmatic scale divided into the factors perspicuity and dependability, whereby two items of the scale efficiency loaded on perspicuity and the other two on dependability.

⁶About the same correlation coefficients were observed for partial correlations between pragmatic quality and attractiveness, $r = .79$, $p < .001$ (controlled for hedonic quality) and between hedonic quality and attractiveness, $r = .60$, $p < .001$, (controlled for pragmatic quality). However, the partial correlation between hedonic and pragmatic quality (controlled for attractiveness) reversed, $r = -.27$, $p < .001$.

Table 4: Correlations between UEQ scales and SUS.

	Per.	Eff.	Dep.	Sti.	Nov.	Att.	Prag.	Hed.
SUS	.78	.77	.74	.53	.29	.74	.83	.45

Note: All correlations are statistically significant ($p < .001$)

includes pragmatic and hedonic aspects as well, a medium sized correlation with the SUS was expected. These hypotheses should hold true independently of the product to be evaluated. Thus, all datasets were included into the analysis, in which participants filled in the SUS in addition to the UEQ ($N = 731$ datasets). In line with the hypotheses, pragmatic aspects of UX highly correlated with the perceived usability of a product, $r = .74$ to $r = .78$, $p < .001$, whereas the correlations between hedonic aspects and usability was small, $r = .29$ to $r = .53$, $p < .001$. The correlation between attractiveness and the SUS was higher than expected, $r = .74$, $p < .001$ (Table 4).

3 STUDY 2: VALIDATION

In this study, we introduced an experimental manipulation of product features in order to evaluate the sensitivity of the UEQ scales to pragmatic and hedonic aspects of product quality. This distinction has a valid theoretical basis in UX research. For example, Carroll and Thomas distinguished the concept *easy to use* from the concept *fun to use* [9]. Logan [24] developed a two-component usability concept that consists of *behavioral* and *emotional* usability. Emotional usability refers "to the degree to which a product is desirable or serves a need(s) beyond the traditional functional objective" (p. 61). Hassenzahl [20] points out that *hedonic* and *ergonomic* quality aspects determine a software's appeal. He defines *ergonomic* qualities as dimensions "that are related to traditional usability, i.e., efficiency and effectiveness" (p.202) and *hedonic* quality as dimensions "with no obvious relation to the task the user wants to accomplish with the system, such as originality, innovativeness, beauty etc." (p.202).

We applied the following relationship between pragmatic and hedonic aspects to manipulate product features, e.g., [3, 20]. Given a very simple user interface, a user should be able to interact effectively and efficiently with it (high pragmatic quality). But this may lead to a boring software system (low hedonic quality). Hedonic quality aspects are supposed to be non-task related. Adding a feature that is unrelated to the task, e.g., color, graphics, or music [25], increases hedonic quality without or with only small impact to pragmatic quality. This holds true, however, only if software complexity is unaffected [3, 20]. If software complexity increases, pragmatic quality decreases.

To allow maximum comparability between products, we designed four apps with a similar purpose (i.e., recording ambient noise) but different user interfaces (see [7] for details). The *Plain* app was kept as simple as possible. Only a single button was provided to record ambient noise (high pragmatic but low hedonic quality). This app was used as a reference for the following experimental manipulations. First, we added a non-task related tutorial (*Plain+Tutorial*). In this tutorial, the correct measurement process was explained. It did not give any information about how to use the app. We hypothesized that scrolling through a nicely designed tutorial should

motivate participants and increase their interest in the task at hand (increased hedonic quality; Hypothesis 1). Second, we added features to the *Plain* app to improve software complexity (*Guided* app). In particular, participants must fulfill some preconditions before they were able to press the record button (e.g., keep their smartphone in a plain position). Because these features were not explained to the participants, we hypothesize that the interaction became less effective and efficient (decreased pragmatic quality, Hypothesis 2). Third, we combined both manipulations simultaneously (*Guided+Tutorial*). A tutorial explained the correct measurement process and how to use the app. Technical features supported the proper implementation by the user. We hypothesized that adding technical features that are understood by the participants reduce software complexity (no decrease of pragmatic quality). Instead, it might be perceived as innovative and stimulating (similar pragmatic quality with increased hedonic quality, Hypothesis 3).

3.1 Methods

Participants. A total of 535 participants volunteered in the study (sample of convenience). Participants with missing data (i.e., not all questions of the UEQ have been answered) were excluded. Thus, $N = 499$ participants remained in the sample (193 women, 304 men; 2 missing values, mean age: 30 years, age range: 18-76 years). All participants gave their written informed consent. The education level of the participants was rather high. About one third had a university degree. About half of the participants ($N = 256$) had a technical or natural scientific background, the others worked in commercial ($N = 72$), medical ($N = 23$), juristic ($N = 6$), pedagogic ($N = 50$), administrative ($N = 13$), manual ($N = 20$) or artistic ($N = 17$) occupational areas. Forty-two participants worked either in other branches ($N = 36$) or did not indicate their line of business ($N = 6$). The study was carried out in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki.

Apps. For this study, we developed four functional variants (flavors) of a mobile app that enabled recording ambient noise with a smartphone (see [7] for details). The apps were run on more than

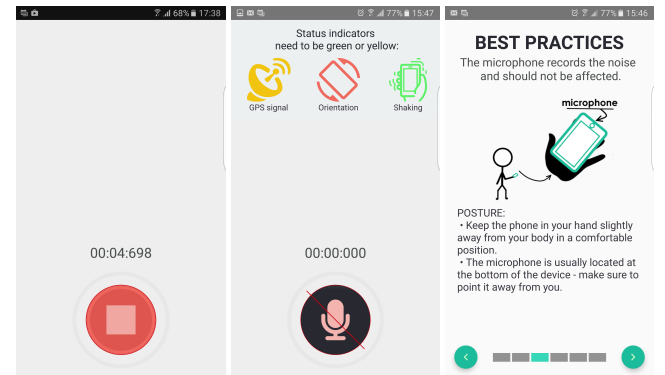


Figure 2: Screenshots of the *Plain* (left) and *Guided* app (middle). Recording is disabled in *Guided* unless certain constraints are met. Additionally, each '+'-version features an in-app-tutorial (right).

100 different smartphones to cover differences in hardware and software configurations. All apps provided a user interface but no additional functionality, such as audio data processing, submitting data to a server, or playing the recorded sound (see Figure 2 for screenshots). The four apps mainly differed in (a) the way participants were instructed on how to record an audio signal properly and (b) the technical guidance for correct recording. The *Plain* app was very minimalist with only one button for starting and stopping recording ambient noise. The same button was also present in the *Guided* app but some technical features were added to guide the measurement process. In particular, the display flipped upside-down to force the user to rotate the smartphone so that the microphone was directed to the to-be-recorded noise. Furthermore, status icons signaled when no GPS signal was available, when the smartphone was not being held in a plain position, and when the user shook the smartphone too strongly. Recording was only possible if these conditions were fulfilled. Whether these technical features support a correct measurement process is out of scope of this paper. Two further apps were identical to those two apps but included a tutorial about how to measure ambient noise properly (*Plain+Tutorial* and *Guided+Tutorial*).

Procedure. At the beginning of the experiment, participants were introduced to the scenario. They were told to imagine that they were part of a community that tries to build a noise map of their city. For that purpose, they would use an app on a smartphone to record the ambient noise level of their environment with the goal to achieve data quality as high as possible. Participants were then asked to record ambient noise levels with a quality as good as possible. They were given one of the recording apps, which they could use as long as they wanted and undertake so many trials until they felt that they recorded a good signal. No further instruction about the measurement process or the handling of the app was given. After completing the task, they filled in the User Experience Questionnaire (UEQ), the System Usability Scale (SUS), and gave qualitative feedback to evaluate the app. In addition, they answered some questions about their smartphone usage behavior as well as a few questions concerning demographics and their habits regarding technology use. The whole experiment took about 15-20 minutes per participant.

3.2 Results

Inter-correlations. As in the first study, the scales measuring pragmatic qualities were highly correlated as well as the scales measuring hedonic quality. Table 5 summarizes the results.

Factor Structure. As in the first study, we run a PCA to evaluate the factor structure of the UEQ. Data quality of the sample was sufficient to run the analysis (KMO=.878, Bartlett test: $\chi^2 = 3961, df = 190, p < .001$). The first analysis was restricted to 5 factors (as claimed by the number of sub-scales of the UEQ), the second analysis extracted 2 factors (as hypothesized by our first study). The results of both analyses are presented in Table 6. The eigenvalues are shown in Figure 3. The five-factor solution explained about 62% of the total variance, which did not substantially differ from those of the first study. We observed some deviations from the expected item loadings. The factors that represent the pragmatic aspects of product quality, i.e., the scales *perspicuity*,

Table 5: Inter-correlations of the UEQ scales (Study 2).

	Att.	Per.	Eff.	Dep.	Sti.	Nov.
Attractiveness (Att.)	–	.37	.43	.47	.76	.54
Perspicuity (Per.)	.37	–	.54	.54	.15	.01
Efficiency (Eff.)	.43	.54	–	.49	.22	.05
Dependability (Dep.)	.47	.54	.49	–	.26	.06
Stimulation (Sti.)	.76	.15	.22	.26	–	.66
Novelty (Nov.)	.54	.01	.05	.06	.66	–

Note: As expected, attractiveness was correlated with all other scales. But there were also great correlations between the pragmatic (green) and hedonic (orange) scales. Coefficients greater than $r = .30$ are statistically significant.

dependability, and *efficiency* were extracted as expected. However, items that were supposed to load on *novelty* and *stimulation* shared a common factor (i.e., hedonic quality aspects), presumably because of the high correlation between *novelty* and *stimulation* (c.f. Table 5). This effect may result from the limited stimulus material. The two-factor solution explained about 46% of the total variance, which is about 10% less than in the first study. Factor loadings were observed as expected, except for a minor loading of Item 11 ("supportive").

Reliability. The reliability ranged between Cronbach's $\alpha = .64$ (scale *efficiency and dependency*) to Cronbach's $\alpha = .89$ (scale *attractiveness*), replicating the results of the previous study.

Hypotheses test. The hypotheses were tested with unpaired t tests. Because of multiple comparisons, p values were adjusted according to Bonferroni. The results are displayed in Table 7.

H1: Manipulating the effects on hedonic quality (*Plain* vs. *Plain+Tutorial*): The *Plain* app with only one button can be used intuitively. As expected, a tutorial that did not provide any information about how to use the app did not affect pragmatic quality, $t(242) = 0.070, p = 1.0$. However, the perceived hedonic quality improved significantly, $t(242) = 5.143, p < .001$. Scrolling through a nicely designed tutorial seems to motivate participants and increases their interest in the task at hand.

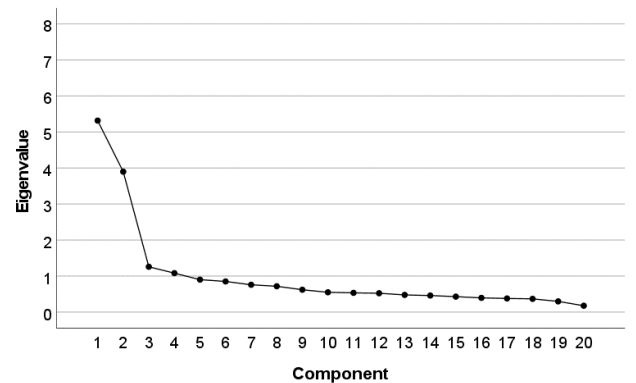


Figure 3: Scree plot of the PCA (Study 2).

Table 6: Factor loadings of the rotated component matrix (Study 2).

Item No.	Description	Orig. Scale	5 Factors with Varimax					2 Factors with Varimax	
			Pers.	Eff.	Dep.	Stim./Nov.	?	Pragmatic	Hedonic
Item 2	understandable	Pers.	.699		.407			.672	
Item 4	easy to learn	Pers.	.801					.663	
Item 13	easy	Pers.	.810					.703	
Item 21	clear	Pers.	.578					.678	
Item 9	fast	Eff.		.525				.553	
Item 20	efficient	Eff.		.647				.629	
Item 22	practical	Eff.		.738				.438	
Item 23	organized	Eff.	.420				.583	.562	
Item 8	predictable	Dep.			.703			.575	
Item 17	secure	Dep.			.530			.473	
Item 19	meet expectations	Dep.			.791			.593	
Item 11	supportive	Dep.			.464	.401		.532	.409
Item 5	valuable	Stim.				.684			.690
Item 6	exciting	Stim.				.807			.800
Item 7	interesting	Stim.				.818			.817
Item 18	motivating	Stim.				.666			.683
Item 3	creative	Nov.				.774			.764
Item 10	inventive	Nov.				.757			.754
Item 15	leading edge	Nov.				.728			.731
Item 26	innovative	Nov.				.735			.739
Eigenvalue			2.600	1.796	2.100	4.797	1.163	4.383	4.832
% Variance explained			13.001	8.979	10.499	23.986	5.815	21.914	24.159

Note: Items are sorted by the original scales, item numbers represent the order in the questionnaire. Unexpected item loadings are highlighted; factor loadings <.400 are omitted; Pers.: Perspicuity, Dep: Dependability, Stim.: Stimulation, Nov.: Novelty

H2: Manipulating the effects on pragmatic quality (*Plain* vs. *Guided*): As expected, pragmatic quality decreased significantly if actions were restricted technically without explanation to the user, $t(238) = 4.843, p < .001$. Interestingly, hedonic quality also increased significantly, $t(238) = 6.074, p < .001$. This result was unexpected because we hypothesized that users might get frustrated. However, the new technical features also added graphics and colors to the display [25]. In particular, the direct feedback for actions might increase the fun when interacting with the app. However, this interpretation remains speculative at this point.

H3: Manipulating the effects on both quality aspects simultaneously (*Plain* vs. *Guided+Tutorial*): As predicted, the app was perceived as efficient and effective to solve the task. The pragmatic quality was similar to the one-button app, $t(243) = 0.173, p = 1.0$. The hedonic quality increased, $t(243) = 7.709, p < .001$.

Table 7: Descriptive results.

Scale	<i>Plain</i>	<i>Plain+Tutorial</i>	<i>Guided</i>	<i>Guided+Tutorial</i>
Pragmatic	1.42	1.42	0.89	1.40
Hedonic	-0.84	0.06	0.03	0.26

Note: Values can vary between -3 and 3.

4 DISCUSSION AND CONCLUSION

In summary, we found (i) acceptable to good reliability of the scales in both studies, as has been previously reported [23, 27]. (ii) We were able to replicate five independent scales (in addition to the scale attractiveness) in the first study but not in the second. The data indicate that a two factor solution which combines the scales *dependability*, *efficiency*, and *perspicuity* to one scale (pragmatic quality) as well as *stimulation* and *novelty* to another one (hedonic quality) is more stable. Thus, not only the reliability further improves but also the fit to the underlying theoretical model. (iii) The second study demonstrated that the scales were sensitive to experimental manipulations of pragmatic and hedonic quality aspects.

4.1 Practical Implications

The reliability of UEQ scales ranged between acceptable and good, which is sufficient for practical use. The two factor solution was very stable. Therefore, we suggest to interpret the overarching factors *perceived pragmatic* and *perceived hedonic quality* to distinguish usability from broader aspects of user experience, at least in a research context. The additional scale *attractiveness* combines both aspects to a general valence impression of the product in terms of like / dislike and complements the UX measurement. Although the original six scales were highly correlated, the interpretation in terms of “educated guesses about the areas where improvements will have the highest impact”(p.6) [23] might still be helpful in industrial

projects. It should be noted, however, that the five aspects can probably not be adapted independently of each other in development and the interpretation should be done only with care. Recently, a German short version of the UEQ has been published that addresses the suggestion to focus on attractiveness, hedonic and pragmatic aspects of UX [30].

4.2 Theoretical Implications

Current theories of UX can be divided into two major classes [31]. *Holistic* theories claim at explaining UX in a comprehensive way. However, this may include constructs that are not operationalizable easily and may end in problems with measuring UX. In contrast, *reductionist* theories focus on the most important aspects of UX and address only those aspects that can be measured.

One of the reductionist models is the one proposed by Hassenzahl [15], which was the basis for the construction of the UEQ. This model distinguishes between perceived ergonomic quality⁷, perceived hedonic quality, and perceived attractiveness of a product, whereby pragmatic and hedonic quality are supposed to be independent dimension of UX. Good ergonomic quality is perceived if a product supports the user effectively and efficiently in task completion. In this sense, ergonomic quality is related to the usability of the product. Hedonic quality results from the user's evaluation with regard to the intrinsic and extrinsic value of the product. Both qualities address human needs. Ergonomic quality aspects fulfill needs of security, control, and trust, whereas hedonic quality fulfills needs of curiosity and social comparison [8]. The current study shows that the UEQ measures these three aspects of UX validly and reliably. However, more detailed information about the factors that contribute, in particular, to hedonic and ergonomic quality cannot be assessed with the UEQ.

It should also be noted that the UEQ does not assess other aspects of UX that are described in alternative theoretical models, e.g., the Component of User Experience model (CUE model [32]). This model consists of three components of UX, (i) the perception of instrumental qualities (such as controllability, effectiveness, learnability etc.), (ii) perceptions of non-instrumental quality (such as visual aesthetics, haptic quality, identification, etc.), and (iii) emotional reactions (i.e., subjective feelings, motor expressions or physiological reaction) that result from both quality aspects. The first two factors are resembled by pragmatic quality (i.e., usability) and hedonic quality of the UEQ, whereas emotional reactions are not part of the questionnaire so far.

4.3 Limitations and Future Work

Before strong conclusion can be drawn, some limitations of the current research should be noted. First, one may argue that the products we used were not complex enough to assess all aspects of UX. However, independent of the maturity of a product, users always perceive attractiveness, pragmatic, and hedonic quality. The maturity level might limit the variance of the evaluative judgments, leading to only small correlations between the scales. But this was not the case in the current study.

Second, product selection was motivated by pragmatic reasons in the first study. Previous data sets were selected based on ecological validity, e.g., different tasks, devices, products, or test environments. This procedure may have led to imbalanced data nevertheless. For example, sample sizes varied. It should be noted, however, that the results of the PCA show that the data fit to a five-component solution in general. Only some items did not load on the expected factor. This might rather be an issue of semantics than on product selection.

Third, it should be noted that evaluating the reliability and validity of the UEQ does not directly contribute to the development of a theoretical model of UX but rather assessed whether the current theoretical models of UX are reflected in the instrument. However, developing a new model was out of scope of this work.

And forth, because we did not apply different instruments to measure UX, we cannot suggest an instrument as being the most appropriate one to measure UX. Again, this was out of scope of this work.

4.4 Conclusion

In this paper, we analyzed the psychometric properties of the User Experience Questionnaire (UEQ). The results show that the UEQ can measure hedonic and pragmatic quality aspects as well as an overall attraction of a product in a reliable and valid way. However, further improvement is needed to measure more detailed facets of these three aspects.

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⁷The term *pragmatic* quality [23] is used synonymously to the term *ergonomic* quality in the model by Hassenzahl [15].

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